REQUIREMENT ANALYSIS — ARTIFACT 05

Solution Requirements Specification

Functional & Non-Functional System Requirements

1. FUNCTIONAL REQUIREMENTS (FR)

Req ID	Module	Requirement Description	Priority
FR-01	Guest Mobile Portal	Support keyless room entry via encrypted Bluetooth BLE tokens.	**High (MVP)**
FR-02	Reservation & CRM	Maintain guest profile history, dietary preferences, and billing preferences across visits.	**High (MVP)**
FR-03	Housekeeping IoT	Automatically assign clean-up tasks based on real-time room motion sensors and guest checkout.	**Medium**
FR-04	Billing & POS Sync	Aggregate charges instantly from fine dining, spa, and golf resort outlets to central room folios.	**High (MVP)**

2. NON-FUNCTIONAL REQUIREMENTS (NFR)

Req ID	Category	Target Metric / Standard	Compliance
NFR-01	Performance	API response time under 200ms for keyless unlock and search operations.	SLA Guaranteed
NFR-02	Security & Privacy	Full PCI-DSS Compliance Level 1 for payment handling and GDPR compliance for guest data.	Mandatory
NFR-03	Availability	99.99% system uptime backed by multi-region cloud deployment.	High Availability
NFR-04	Scalability	Support peak traffic bursts up to 5,000 concurrent active guest and staff sessions.	Auto-scaling